IGCSE 0607 Practice Questions — E3.7 — Combined Questions & Markscheme

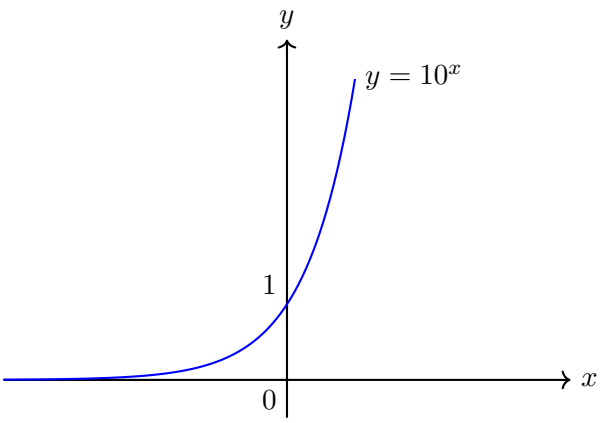
The logarithmic function

Total Questions: 39 — **Total Marks:** 235 Marks

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- **E3.7** The logarithmic function
  → **Paper 2 (Non-Calculator)** *(13 Qs)* · **Paper 4 (Calculator)** *(26 Qs)*

1 The diagram shows a sketch of the function $f(x) = 10^x$.



(a) Write down the inverse function, $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

(b) Work out the exact value of $f^{-1}\left(\frac{1}{100}\right)$.

..... **[1]**

(c) Write down the exact solution to the equation $f(x) = 45$.

$x =$ ..... **[1]**

2 **(a)** Write $10^3 = 1000$ as a logarithmic equation.

..... **[1]**

(b) Find the exact value of x when $\log x = -2$.

$x =$ ..... **[1]**

(c) Solve $5^x = 12$, giving your answer in the form $\dfrac{\log a}{\log b}$.

$x =$ ..... **[1]**

3 The functions f and g are defined as $f(x) = 10^x$ and $g(x) = \log x$.

(a) Find the value of $g(100)$.

..... **[1]**

(b) Find the value of x when $f(x) = 0.001$.

$x =$ ..... **[1]**

(c) Write down the expression for $g(f(x))$ in its simplest form.

$g(f(x)) =$ ..... **[1]**

4 **(a)** Work out the value of

$\log 10000$

..... **[1]**

(b) Write the following equation in the form $\log_a b = c$.

$5^3 = 125$

..... **[1]**

(c) Solve the equation

$\log_2 x = 6$

$x =$ ..... **[1]**

5 **(a)** Find the value of x when $\log_2 x = 5$.

$x =$ ..... **[1]**

(b) Find the value of y when $10^y = 0.01$.

$y =$ ..... **[1]**

(c) Write the equation $P = k^t$ in logarithmic form.

..... **[1]**

6 **(a)** Write the following as a logarithmic equation.

$5^3 = 125$

..... **[1]**

(b) Find the value of $\log 1000$.

..... **[1]**

(c) Solve the equation.

$\log_2 x = 5$

$x =$ ..... **[2]**

7 **(a)** Write the equation $10^5 = 100000$ in the form $\log a = b$.

..... **[1]**

(b) Find the value of x when $\log x = 3$.

$x =$ ..... **[2]**

(c) The function $f(x)$ is defined as $f(x) = 10^x$.
Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

8 All logarithms in this question are in the base shown, or base 10
10 where no base is written.

(a) Write down the value of

(i) log
1000
log1000,

..... **[1]**

(ii) log
2
32
log
2
32.

..... **[1]**

(b) Given that
log
3
=
4
log
3
x=4, find the value of
x.

=
x= ..... **[1]**

(c) The function
f is defined by
(
)
=
10
f(x)=10
x
.
Write down
1
(

)
f
1
(x).

1
(
)
=
f
1
(x)= ..... **[1]**

9 **(a)** Find the value of x in each of the following equations.

(i) $10^x = 100000$

$x =$ ..... **[1]**

(ii) $\log x = -2$

$x =$ ..... **[1]**

(iii) $\log_2 64 = x$

$x =$ ..... **[1]**

(b) Given that $f(x) = 10^x$, write down an expression for $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

10 **(a)** Write down the value of

$\log 1000$

..... **[1]**

(b) Solve the equation.

$\log x = -2$

$x =$ ..... **[2]**

(c) Write the equation $5^x = 125$ in the form $x = \log_a b$.

$x =$ ..... **[1]**

11 Answer the following independent questions on logarithmic and exponential equations.

(a) Write the exponential equation $10^4 = 10000$ in logarithmic form.

..... **[1]**

(b) Find the exact value of x when $\log_5 x = 3$.

$x =$ ..... **[2]**

(c) The solution to the equation $7^x = 25$ can be written in the form $x = \dfrac{\log b}{\log a}$.
Write down the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

12 Consider the function $f(x) = \log_3 x$ for $x > 0$.

(a) Find the value of $f(81)$.

..... **[1]**

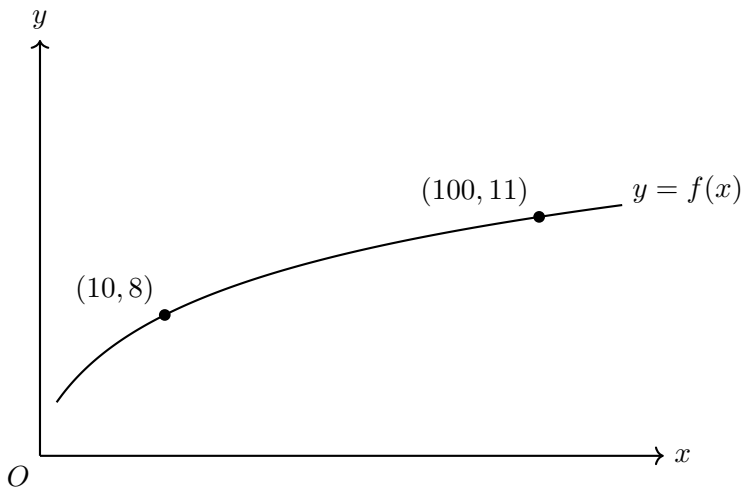
(b) Find the value of $f\left(\frac{1}{9}\right)$.

..... **[2]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

13 The diagram shows a sketch of the graph of $y = f(x)$ for $x > 0$.
The function is defined as $f(x) = p\log x + q$, where $\log x$ denotes the logarithm to base 10.
The graph passes through the points $(10, 8)$ and $(100, 11)$.



(a) **(a)** Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[3]**

(b) Solve the equation $f(x) = 23$.

$x =$ ..... **[2]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

14 The logarithmic function is the inverse of the exponential function.

(a) Write the equation $y = 6^x$ in logarithmic form.

(a) Write the equation $y = 6^x$ in logarithmic form.

$x =$ ..... **[1]**

(b) Solve the equation.

$3^x = 20$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

15 A population of insects is growing exponentially.
The number of insects, N, after t days is given by

$$N = 250 \times 1.08^t$$

(a) Work out the number of insects after 10 days.
Give your answer to the nearest integer.

..... **[2]**

(b) Calculate the number of days it takes for the population to reach 1000.
Give your answer correct to 1 decimal place.

..... days **[3]**

16 The value of a machine, V dollars, depreciates exponentially according to the formula

$$V = 15000 \times 0.8^t$$

where t is the number of years after it is purchased.

(a) Calculate the value of the machine after 4 years.

$ ..... **[2]**

(b) Find the number of years it takes for the value of the machine to decrease to $7500.
Give your answer correct to 3 significant figures.

..... years **[3]**

17 A machine is purchased for \$25000. Its value depreciates exponentially by 12% each year.

(a) Calculate the value of the machine after 4 years.
Give your answer correct to the nearest dollar.

% ..... **[2]**

(b) Find the number of complete years it takes for the value of the machine to fall below \$10000.
You must show all your working.

..... years **[3]**

18 **(a)** Solve the equation.

$3^{2x} = 50$ Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

(b) Make x the subject of the formula.

$y = 10^{x-4}$

$x =$ ..... **[2]**

19 A population of bacteria is modelled by the formula

$P = 5000 \times 1.12^t$ where P is the population and t is the time in hours after the start of an experiment.

(a) **(a)** Write down the initial population of the bacteria.

..... **[1]**

(b) Find the time it takes for the population to reach 20 000.
Give your answer correct to 1 decimal place.

$t =$ ..... hours **[4]**

20 An investment of \$8000 earns compound interest at a rate of 3.5% per year. The value of the investment after n years is \$$V$.

(a) Write down an equation for V in terms of n.

$V =$ ..... **[1]**

(b) Calculate the minimum number of complete years it takes for the value of the investment to exceed \$12000.
Show all your working.

..... years **[3]**

(c) Solve the equation.

$2^x = 55$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

21 **(a)** Solve the equation $3^x = 20$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

(b) The mass, M grams, of a radioactive substance after t days is given by the formula
$M = 500 \times 0.85^t$ Calculate the value of t when $M = 100$.
Give your answer correct to 3 significant figures.

$t =$ ..... **[3]**

22 The value, V dollars, of a painting t years after it was bought is given by the formula

$$V = 4500 \times 1.08^t$$

(a) Calculate the value of the painting 5 years after it was bought.

Give your answer correct to the nearest dollar.

..... **[2]**

(b) Calculate the time it takes for the value of the painting to reach \$10 000.

Give your answer in years, correct to 2 decimal places.

..... years **[4]**

23 The value, $\$V$, of a vintage motorcycle t years after it was purchased is modelled by the formula

$$V = 4500 \times 1.08^t$$

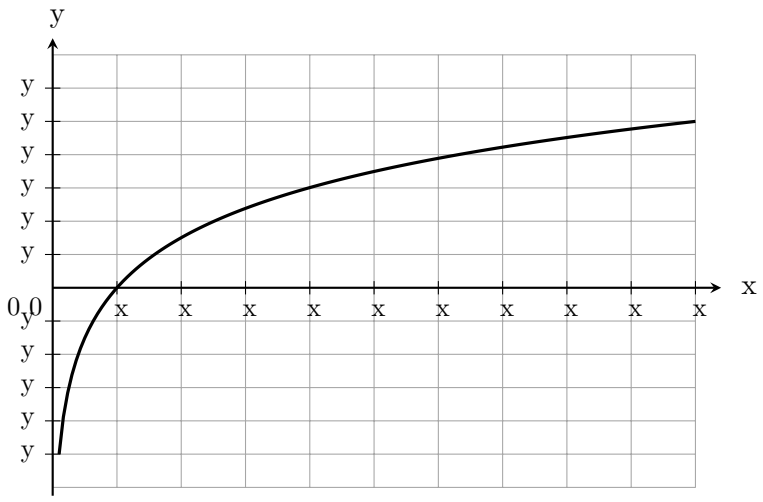
(a) Calculate the value of the motorcycle after 5 years.

$\$$ ..... **[2]**

(b) Calculate the number of years it takes for the value of the motorcycle to reach \$10 000. Give your answer correct to 1 decimal place.

..... years **[4]**

24 The diagram shows the graph of
=
log
y=logx for
0.1
10
0.1x10.



(a) Use the graph to estimate the value of
log
6
log6.

..... **[1]**

(b) Use the graph to solve the equation
log
=
0.5
logx=0.5.

=
x= ..... **[1]**

(c) Solve the equation
3
+
1
=
20
3
x+1
=20, giving your answer correct to

3
3 significant figures.

=
x= ..... **[3]**

(d) Solve the equation
log
(
2
3
)
=
1.4
log(2x3)=1.4, giving your answer correct to
3
3 significant figures.

=
x= ..... **[2]**

25 The mass, M grams, of a radioactive substance after t days is given by the formula $M = 400 \times 0.88^t$.

(a) Calculate the mass of the substance after 5 days.
Give your answer correct to 1 decimal place.

..... grams **[2]**

(b) Find the number of days it takes for the mass of the substance to reduce to exactly half of its initial value.

..... days **[3]**

(c) Another radioactive substance has an initial mass of 800 grams.
After 10 days, its mass is 300 grams.
It decays according to the formula $M = 800 \times k^t$.
Find the value of k, giving your answer correct to 3 significant figures.

$k =$ ..... **[3]**

26 An amount of \$8000 is invested in an account that pays compound interest at a rate of 4.5% per year. The value of the investment, V dollars, after t years is given by the formula

$$V = 8000 \times 1.045^t$$

(a) Calculate the value of the investment after 5 years.

\$ ..... **[2]**

(b) Use logarithms to calculate the number of complete years it takes for the value of the investment to first exceed \$15000.

..... years **[3]**

(c) Solve the equation.

$7^{x-1} = 250$ Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

27 A population of insects grows exponentially.
The number of insects, N, after t weeks is given by the formula

$$N = 800 \times 2^t$$

(a) Work out the number of insects after 4 weeks.

..... **[1]**

(b) The number of insects reaches 102 400.
Work out the exact value of t.

$t =$ ..... **[2]**

(c) Another population of insects can be modelled by the formula $P = 1500 \times 3^t$.
Find the value of t when this population reaches 20 000.
Give your answer correct to 3 significant figures.

$t =$ ..... **[3]**

(d) A sum of \$2000 is invested at an annual compound interest rate of 5%. The total value, V, of the investment after n years is given by

$$V = 2000 \times 1.05^n$$

Calculate the minimum number of complete years required for the investment to double in value.

..... years **[3]**

28 A function f is defined as $f(x) = 4 \times 10^{2x} - 7$.

(a) Find $f^{-1}(x)$.

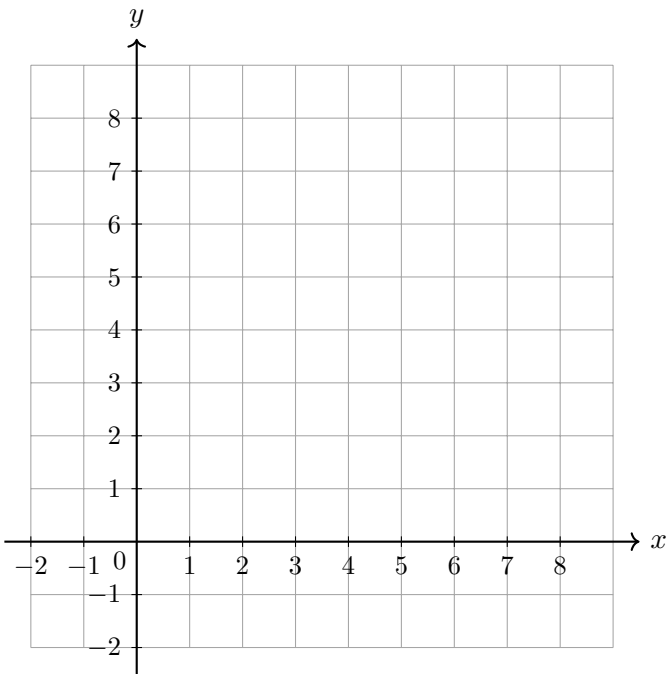
$f^{-1}(x) =$ ..... **[3]**

(b) Solve the equation $f(x) = 53$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

29 A function is defined as $f(x) = 2^x$.

(a) On the grid, sketch the graph of $y = f(x)$ for $-2 \leq x \leq 3$.



[2]

(b) Write down the equation of the inverse function, $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

(c) On the same grid in part (a), sketch the graph of $y = f^{-1}(x)$ for $0 < x \leq 8$. **[2]**

(d) Solve the equation $2^x = 12$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

30 **(a)** The mass, M grams, of a radioactive substance after t days is given by $M = 250 \times 0.92^t$.

(i) Write down the initial mass of the substance.

..... grams **[1]**

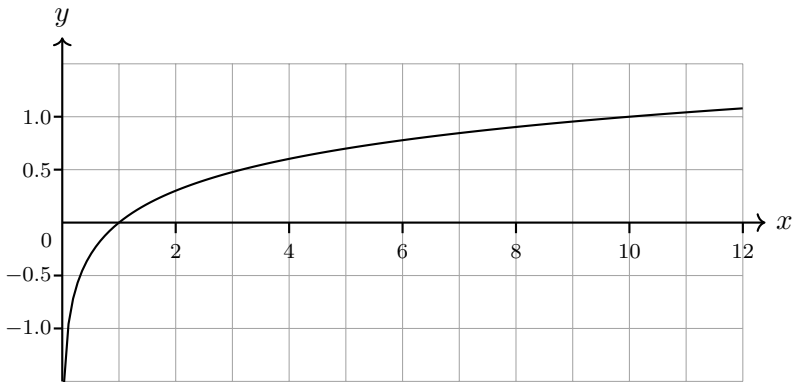
(ii) Find the number of days it takes for the mass to reduce to 50 grams.
Give your answer correct to 3 significant figures.

..... days **[3]**

(b) An amount of \$800 is invested at a rate of 6% per year compound interest. Calculate the number of complete years it takes for the investment to first exceed \$2000.

..... years **[3]**

31 The diagram shows the graph of $y = \log x$ for $0 < x \leq 12$, where $\log x$ denotes $\log_{10} x$.



(a) Write down the coordinates of the x-intercept and the equation of the vertical asymptote of the graph of $y = \log x$.

x-intercept = ... , Asymptote: ..... **[2]**

(b) Use the graph to estimate the solution to the equation $\log x = 0.7$.

x = ... **[1]**

(c) Solve algebraically the equation

$$3^{2x-1} = 5^{x+2}$$

Give your answer correct to 3 significant figures.

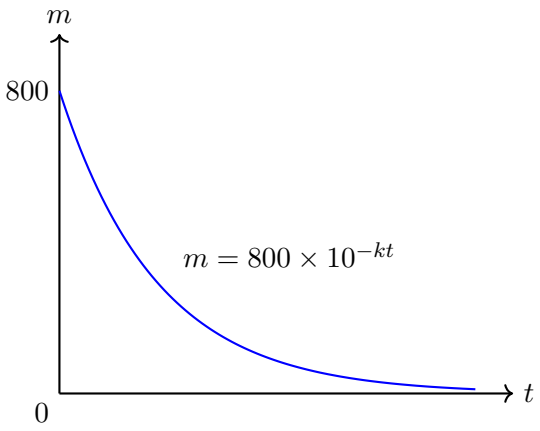
x = ... **[4]**

32 The mass, m grams, of a radioactive isotope decays according to the formula

$m = 800 \times 10^{-kt}$ where t is the time in years and k is a constant.

The diagram shows a sketch of the decay curve.

After 15 years, the mass is 250 grams.



(a) Show that $k = 0.0337$ correct to 3 significant figures.

[3]

(b) Find the number of years it takes for the mass to become 50 grams.

..... years **[3]**

(c) Make t the subject of the formula $m = 800 \times 10^{-kt}$.

$t =$ ..... **[2]**

33 Let $f(x) = 5 \times 10^x$ and $g(x) = \log x$.

(a) Find the value of $f(g(100))$.

..... **[2]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Solve the equation.

$f(x) = 200$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

(d) Find $g(f(x))$ in the form $x + k$, where k is a constant.

..... **[2]**

34 A function is defined as $f(x) = 5 \times 3^x - 2$.

(a) Find the value of $f(-2)$, giving your answer as a fraction in its simplest form.

..... **[2]**

(b) Find $f^{-1}(x)$.

Give your answer in terms of base 10 logarithms.

$f^{-1}(x) =$ ..... **[3]**

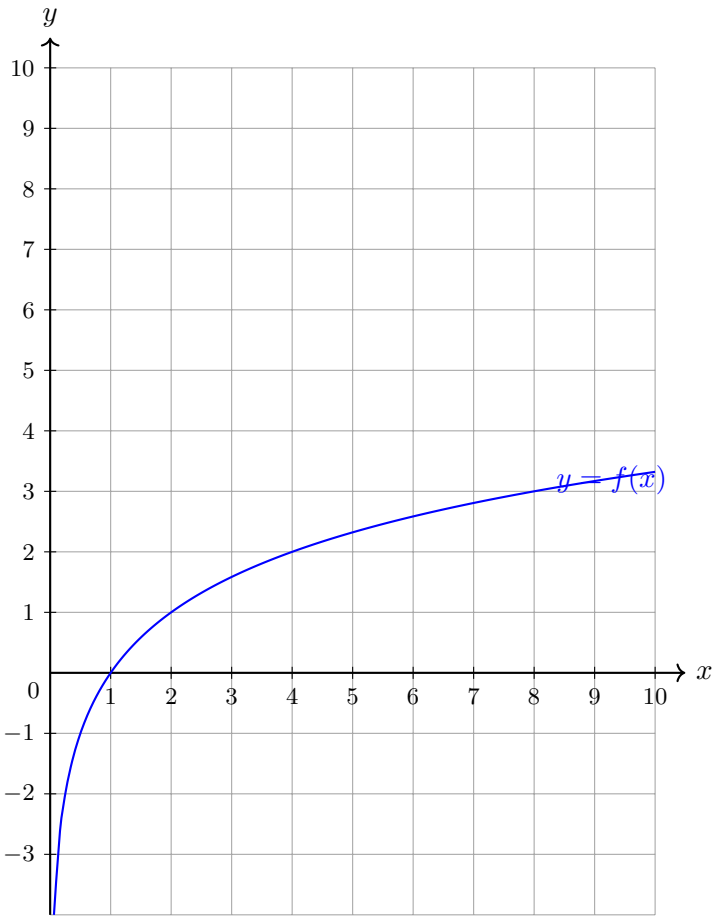
(c) Solve the equation.

$f(x) = 500$

Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

35 The diagram shows the graph of $y = f(x)$.



(a) The graph of $y = f(x)$ passes through the point $(8, 3)$.
Given that $f(x) = \log_a x$, show that $a = 2$.

[2]

(b) **(i)** On the grid, sketch the graph of $y = f^{-1}(x)$. **[2]**

(ii) Write down the equation of the asymptote of the graph of $y = f^{-1}(x)$.

..... **[1]**

(c) By drawing a suitable straight line on the grid, find an approximate solution to the equation $\log_2 x + x = 8$.

$x =$ ..... **[3]**

36 The function $f(x)$ is defined as $f(x) = 4 \times 10^{0.5x}$.

(a) Find the value of $f(3)$.
Give your answer correct to 3 significant figures.

..... **[2]**

(b) Find $f^{-1}(x)$.

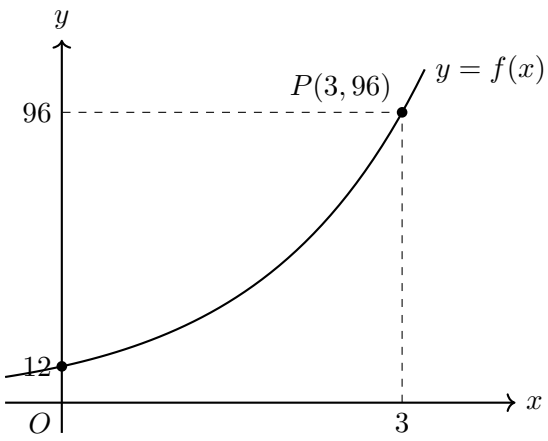
$f^{-1}(x) =$ ..... **[3]**

(c) Solve the equation

$$f(x) = 200$$

$x =$ ..... **[3]**

37 The diagram shows the graph of $y = f(x)$ where $f(x) = a \times b^x$.
The graph intersects the y-axis at the point $(0, 12)$ and passes through the point $P(3, 96)$.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

(b) Find $f^{-1}(x)$.
Give your answer in terms of base 10 logarithms.

$f^{-1}(x) =$ ..... **[3]**

(c) Solve the equation $f(x) = 500$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

38 The value, \$V, of a machine t years after it is purchased is modelled by the formula $V = 24000 \times 0.85^t$.

(a) Calculate the value of the machine after 4 years, giving your answer correct to the nearest dollar.

\$ ..... **[2]**

(b) Find the number of years it takes for the value of the machine to fall to \$9000.
Give your answer correct to 3 significant figures.

$t =$ ..... years **[3]**

(c) A different asset increases in value by 6.8% each year.
Find the minimum whole number of years it takes for the value of this asset to double.

..... years **[3]**

(d) The function g is defined by $g(x) = 5 \times 2^x + 3$.
Find $g^{-1}(x)$.

$g^{-1}(x) =$ ..... **[3]**

39 The function $f(x) = A \times b^x$, where $b > 0$, passes through the points $(0, 7)$ and $(3, 56)$.

(a) Find the value of A and the value of b.

$A =$ ..... , $b =$ ..... **[3]**

(b) Find the inverse function, $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(c) The equation $f(x) = 600$ is equivalent to $2^x = k$.

(i) Work out the exact value of k as a fraction.

$k =$ ..... **[1]**

(ii) Solve the equation $f(x) = 600$.
Give your answer correct to 3 decimal places.

$x =$ ..... **[2]**

(d) A different function $g(x)$ is defined as $g(x) = \log_c(x)$.
The graph of $y = g(x)$ passes through the point $(81, 4)$.

(i) Find the value of c.

$c =$ ..... **[2]**

(ii) Solve the equation $g(x) = -2$.

$x =$ ..... **[1]**

1 **(a)** $\log_{10} x$ OR $\log x$ **[B1]**

(b) -2 **[B1]**

(c) $\log_{10} 45$ OR $\log 45$ **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** $\log 1000 = 3$ **[B1]**

(b) 0.01 or $\frac{1}{100}$ **[B1]**

(c) $\frac{\log 12}{\log 5}$ **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** 2 **[B1]**

(b) -3 **[B1]**

(c) x **[B1]**

[Back to Question] *[Total Marks: 3]*

4 **(a)** 4 **[B1]**

(b) $\log_5 125 = 3$ **[B1]**

(c) 64 **[B1]**

[Back to Question] *[Total Marks: 3]*

5 **(a)** 32 **[B1]**

(b) -2 **[B1]**

(c) $t = \log_k P$ **[B1]**

[Back to Question] *[Total Marks: 3]*

6 **(a)** $\log_5 125 = 3$ **[B1]**

(b) 3 **[B1]**

(c) $x = 2^5$ **[M1]**

32 **[A1]**

[Back to Question] *[Total Marks: 4]*

7 **(a)** $\log 100000 = 5$ **[B1]**

(b) 10^3 **[M1]**

1000 **[A1]**

(c) $\log x$ **[B1]**

[Back to Question] *[Total Marks: 4]*

8 **(a)** **(i)** 3
3 **[B1]**

(ii) 5
5 **[B1]**

(b)
=
3
4
=
81
x=3
4
=81 **[B1]**

(c) log
logx (or
log
10
log
10
x) **[B1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** **(i)** 5 **[B1]**

(ii) 0.01 or $\frac{1}{100}$ **[B1]**

(iii) 6 **[B1]**

(b) $\log x$ or $\log_{10} x$ **[B1]**

[Back to Question] *[Total Marks: 4]*

10 **(a)** 3 **[B1]**

(b) 10^{-2} or $\frac{1}{100}$ **[M1]**

0.01 **[A1]**

(c) $\log_5 125$ **[B1]**

[Back to Question] *[Total Marks: 4]*

11 **(a)** $\log_{10} 10000 = 4$ (or $\log 10000 = 4$) **[B1]**

(b) Uses index form $x = 5^3$ **[M1]**

125 **[A1]**

(c) $a = 7$ **[B1]**

$b = 25$ **[B1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** 4 **[B1]**

(b) Writes $\frac{1}{9} = 3^{-2}$ **[M1]**

-2 **[A1]**

(c) Writes $y = \log_3 x$ as $x = 3^y$ (or $y = 3^x$) **[M1]**

3^x **[A1]**

[Back to Question] *[Total Marks: 5]*

13 **(a)** **(a)** $p(1) + q = 8$ or $p(2) + q = 11$ **[B1]**

A valid method to solve simultaneous equations e.g. $(2p + q) - (p + q) = 11 - 8$ **[M1]**

$p = 3$, $q = 5$ **[A1]**

(b) $3 \log x + 5 = 23 \implies 3 \log x = 18 \implies \log x = 6$ **[M1]**

$1\,000\,000$ or 10^6 **[A1]**

(c) $x = 3 \log y + 5 \implies \log y = \dfrac{x - 5}{3}$ **[M1]**

$10^{\frac{x-5}{3}}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

14 **(a)** **(a)** $\log_6 y$ **[B1]**

(b) $x = \dfrac{\log 20}{\log 3}$ or $x \log 3 = \log 20$ **[M1]**

2.73 **[A1]**

[Back to Question] *[Total Marks: 3]*

15 **(a)** 250×1.08^{10} **[M1]**

540 **[A1]**

(b) $250 \times 1.08^t = 1000$ **[M1]**

$1.08^t = 4 \implies t = \dfrac{\log 4}{\log 1.08}$ **[M1]**

18.0 **[A1]**

[Back to Question] *[Total Marks: 5]*

16 **(a)** 15000×0.8^4 **[M1]**

6144 **[A1]**

(b) $7500 = 15000 \times 0.8^t$ **[M1]**

$0.5 = 0.8^t \implies t = \dfrac{\log 0.5}{\log 0.8}$ **[M1]**

3.11 **[A1]**

[Back to Question] *[Total Marks: 5]*

17 **(a)** $25000 \times (1 - 0.12)^4$ or 25000×0.88^4 **[M1]**

14992 **[A1]**

(b) $25000 \times 0.88^n = 10000$ or $0.88^n = 0.4$ **[M1]**

$n = \dfrac{\log 0.4}{\log 0.88}$ **[M1]**

$n = 7.16...$ **[A1]**

8 **[B1]**

[Back to Question] *[Total Marks: 5]*

18 **(a)** $2x = \dfrac{\log 50}{\log 3}$ **[M1]**

$2x = 3.56\ldots$ **[A1]**

1.78 **[A1]**

(b) $\log y = x - 4$ **[M1]**

$\log y + 4$ **[A1]**

[Back to Question] *[Total Marks: 5]*

19 **(a)** **(a)** 5000 **[B1]**

(b) $5000 \times 1.12^t = 20000$ **[M1]**

$1.12^t = 4$ **[M1]**

$t = \dfrac{\log 4}{\log 1.12}$ **[M1]**

12.2 **[A1]**

[Back to Question] *[Total Marks: 5]*

20 **(a)** 8000×1.035^n **[B1]**

(b) Sets up the inequality or equation: $8000 \times 1.035^n > 12000$ OR $1.035^n = 1.5$ **[M1]**

Uses logarithms to solve for n: $n = \dfrac{\log_{10} 1.5}{\log_{10} 1.035}$ **[M1]**

12 **[A1]**

(c) $x = \dfrac{\log_{10} 55}{\log_{10} 2}$ **[M1]**

5.78 **[A1]**

[Back to Question] *[Total Marks: 6]*

21 **(a)** $x \log 3 = \log 20$ **[M1]**

$x = \dfrac{\log 20}{\log 3}$ **[M1]**

2.73 **[A1]**

(b) $100 = 500 \times 0.85^t \implies 0.85^t = 0.2$ **[M1]**

$t \log 0.85 = \log 0.2 \implies t = \dfrac{\log 0.2}{\log 0.85}$ **[M1]**

9.90 **[A1]**

[Back to Question] *[Total Marks: 6]*

22 **(a)** 4500×1.08^5 **[M1]**

6612 **[A1]**

(b) $4500 \times 1.08^t = 10000$ **[M1]**

$1.08^t = \dfrac{10000}{4500}$ oe **[M1]**

$t = \dfrac{\log(20/9)}{\log 1.08}$ **[M1]**

10.38 **[A1]**

[Back to Question] *[Total Marks: 6]*

23 **(a)** 4500×1.08^5 **[M1]**

6611.98 (or 6610 to 3 significant figures) **[A1]**

(b) $4500 \times 1.08^t = 10000$ **[M1]**

$1.08^t = \dfrac{10000}{4500}$ or $1.08^t = \dfrac{20}{9}$ **[M1]**

$t = \dfrac{\log\left(\dfrac{20}{9}\right)}{\log 1.08}$ **[M1]**

10.4 (10.376...) **[A1]**

[Back to Question] *[Total Marks: 6]*

24 **(a)** 0.78
0.78 (accept
0.75
0.75 to
0.80
0.80) **[B1]**

(b) 3.2
3.2 (accept
3.1
3.1 to
3.3
3.3) **[B1]**

(c) (
+
1
)
log
3
=
log
20
(x+1)log3=log20 or
+
1
=
log
20
log
3
x+1=
log3
log20
[M1]

+
1
=
2.7268
...
x+1=2.7268... **[M1]**

1.73
1.73 **[A1]**

(d) 2
3
=
10
1.4
2x3=10
1.4
(or
2
3

=
25.118
...
2x3=25.118...) **[M1]**

14.1
14.1 **[A1]**

[Back to Question] *[Total Marks: 7]*

25 **(a)** Substitutes $t = 5$ into the formula: 400×0.88^5 **[M1]**

211.1 **[A1]**

(b) Sets up the equation $400 \times 0.88^t = 200$ and simplifies to $0.88^t = 0.5$ **[M1]**

$t = \dfrac{\log 0.5}{\log 0.88}$ **[M1]**

5.42 **[A1]**

(c) Sets up the equation $800 \times k^{10} = 300$ **[M1]**

$k^{10} = 0.375$ **[M1]**

0.907 **[A1]**

[Back to Question] *[Total Marks: 8]*

26 **(a)** 8000×1.045^5 **[M1]**

9969.46 (accept 9969 or 9970) **[A1]**

(b) $1.045^t = \dfrac{15000}{8000}$ or 1.875 **[M1]**

$t = \dfrac{\log 1.875}{\log 1.045}$ or 14.28... **[M1]**

15 **[A1]**

(c) $\log(7^{x-1}) = \log 250$ or $x - 1 = \log_7 250$ **[M1]**

$x - 1 = \dfrac{\log 250}{\log 7}$ or $x - 1 = 2.837...$ **[M1]**

3.84 **[A1]**

[Back to Question] *[Total Marks: 8]*

27 **(a)** 12800 **[B1]**

(b) $800 \times 2^t = 102400 \implies 2^t = 128$ **[M1]**

$t = 7$ **[A1]**

(c) $1500 \times 3^t = 20000 \implies 3^t = \dfrac{40}{3}$ (or 13.33...) **[M1]**

Uses logarithms to solve $3^t = \dfrac{40}{3}$, e.g. $t = \dfrac{\log(40/3)}{\log 3}$ **[M1]**

2.36 (from 2.358...) **[A1]**

(d) $2000 \times 1.05^n = 4000 \implies 1.05^n = 2$ **[M1]**

$n = \dfrac{\log 2}{\log 1.05}$ (or 14.2...) **[M1]**

15 **[A1]**

[Back to Question] *[Total Marks: 9]*

28 **(a)** $y = 4 \times 10^{2x} - 7 \implies y + 7 = 4 \times 10^{2x}$ **[M1]**

$\dfrac{y+7}{4} = 10^{2x} \implies 2x = \log\left(\dfrac{y+7}{4}\right)$ **[M1]**

$\dfrac{1}{2}\log\left(\dfrac{x+7}{4}\right)$ **[A1]**

(b) $4 \times 10^{2x} - 7 = 53 \implies 4 \times 10^{2x} = 60$ **[M1]**

$10^{2x} = 15 \implies 2x = \log 15$ **[M1]**

0.588 **[A1]**

[Back to Question] *[Total Marks: 6]*

29 **(a)** Exponential curve passing through $(0, 1)$ and $(1, 2)$ **[B1]**

Curve passing through $(3, 8)$ and approaching the negative x-axis as an asymptote **[B1]**

(b) $\dfrac{\log x}{\log 2}$ (or $\log_2 x$) **[B1]**

(c) Logarithmic curve passing through $(1, 0)$ and $(2, 1)$ **[B1]**

Curve passing through $(8, 3)$ and approaching the negative y-axis as an asymptote **[B1]**

(d) $x = \dfrac{\log 12}{\log 2}$ **[M1]**

3.58 **[A1]**

[Back to Question] *[Total Marks: 7]*

30 **(a)** **(i)** 250 **[B1]**

(ii) $250 \times 0.92^t = 50$ or $0.92^t = 0.2$ **[M1]**

$t = \dfrac{\log 0.2}{\log 0.92}$ **[M1]**

19.3 **[A1]**

(b) $800 \times 1.06^n > 2000$ or $1.06^n = 2.5$ **[M1]**

$n = \dfrac{\log 2.5}{\log 1.06}$ $(= 15.7\ldots)$ **[M1]**

16 **[A1]**

[Back to Question] *[Total Marks: 7]*

31 **(a)** x-intercept $= (1, 0)$ **[B1]**

Asymptote: $x = 0$ **[B1]**

(b) 5.0 (accept 4.9 to 5.1) **[B1]**

(c) Takes logarithms of both sides: $(2x - 1)\log 3 = (x + 2)\log 5$ **[M1]**

Expands and collects x terms: $x(2\log 3 - \log 5) = 2\log 5 + \log 3$ **[M1]**

$x = \dfrac{2\log 5 + \log 3}{2\log 3 - \log 5}$ (or $\dfrac{\log 75}{\log 1.8}$) **[M1]**

7.34 (7.3414...) **[A1]**

[Back to Question] *[Total Marks: 7]*

32 **(a)** Substitutes given values into the formula: $250 = 800 \times 10^{-15k}$ **[M1]**

Rearranges to isolate the exponential: $10^{-15k} = 0.3125$ **[M1]**

Takes logarithms to solve for k: $-15k = \log_{10}(0.3125) \implies k = \dfrac{\log_{10}(0.3125)}{-15} = 0.03367...$ **[A1]**

(b) Substitutes $m = 50$ and $k = 0.03367...$ into the formula: $50 = 800 \times 10^{-0.03367t}$ **[M1]**

Isolates the exponential and takes logarithms: $-0.03367t = \log_{10}\left(\dfrac{50}{800}\right)$ **[M1]**

35.8 (accept 35.7 to 35.8) **[A1]**

(c) Rearranges to $10^{-kt} = \dfrac{m}{800}$ and converts to logarithmic form: $-kt = \log_{10}\left(\dfrac{m}{800}\right)$ **[M1]**

$t = -\dfrac{1}{k}\log_{10}\left(\dfrac{m}{800}\right)$ OR $t = \dfrac{1}{k}\log_{10}\left(\dfrac{800}{m}\right)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

33 **(a)** $g(100) = \log 100 = 2$ **[M1]**

$f(2) = 5 \times 10^2 = 500$ **[A1]**

(b) $y = 5 \times 10^x$ leads to $10^x = \dfrac{y}{5}$ **[M1]**

$\log\left(\dfrac{x}{5}\right)$ or $\log x - \log 5$ **[A1]**

(c) $5 \times 10^x = 200$ leads to $10^x = 40$ **[M1]**

$x = \log 40 = 1.602...$ **[A1]**

1.60 **[B1]**

(d) $g(f(x)) = \log(5 \times 10^x)$ **[M1]**

$= \log 5 + \log 10^x = \log 5 + x$ **[M1]**

$x + \log 5$ **[A1]**

[Back to Question] *[Total Marks: 8]*

34 **(a)** $5 \times 3^{-2} - 2$ **[M1]**

$-\dfrac{13}{9}$ **[A1]**

(b) $y = 5 \times 3^x - 2 \implies y + 2 = 5 \times 3^x$ **[M1]**

$3^x = \dfrac{y+2}{5}$ **[M1]**

$\dfrac{\log\left(\frac{x+2}{5}\right)}{\log 3}$ **[A1]**

(c) $5 \times 3^x - 2 = 500 \implies 3^x = 100.4$ **[M1]**

$x = \dfrac{\log 100.4}{\log 3}$ **[M1]**

4.20 **[A1]**

[Back to Question] *[Total Marks: 8]*

35 **(a)** $3 = \log_a 8$ **[M1]**

$a^3 = 8 \implies a = 2$ **[A1]**

(b) **(i)** Smooth exponential curve passing through $(0, 1)$ **[B1]**

Correct shape passing through $(1, 2)$, $(2, 4)$ and $(3, 8)$ **[B1]**

(ii) $y = 0$ (accept the x-axis) **[B1]**

(c) Rearranges to $\log_2 x = 8 - x$ **[M1]**

Draws the line $y = 8 - x$ correctly on the grid (passing through $(0, 8)$ and $(8, 0)$) **[M1]**

$x = 5.5$ (accept 5.4 to 5.6) **[A1]**

[Back to Question] *[Total Marks: 8]*

36 **(a)** Substitutes $x = 3$: $4 \times 10^{1.5}$ **[M1]**

126 **[A1]**

(b) Sets $y = 4 \times 10^{0.5x}$ and rearranges to $\dfrac{y}{4} = 10^{0.5x}$ **[M1]**

$\log\left(\dfrac{y}{4}\right) = 0.5x$ **[M1]**

$2\log\left(\dfrac{x}{4}\right)$ **[A1]**

(c) $4 \times 10^{0.5x} = 200$

$10^{0.5x} = 50$ [M1]

$0.5x = \log 50$ **[M1]**

3.40 **[A1]**

[Back to Question] *[Total Marks: 8]*

37 **(a)** $a \times b^0 = 12 \implies a = 12$ **[B1]**

$12 \times b^3 = 96 \implies b^3 = 8$ **[M1]**

$b = 2$ **[A1]**

(b) Sets up $y = 12 \times 2^x \implies 2^x = \frac{y}{12}$ **[M1]**

$x \log 2 = \log\left(\frac{y}{12}\right)$ **[M1]**

$f^{-1}(x) = \frac{\log(x/12)}{\log 2}$ **[A1]**

(c) $12 \times 2^x = 500 \implies 2^x = \frac{125}{3}$ (41.666...) **[M1]**

$x = \frac{\log(125/3)}{\log 2}$ **[M1]**

5.38 **[A1]**

[Back to Question] *[Total Marks: 9]*

38 (a) 24000
×
0.85
4
24000×0.85
4
(
=
12528.15
=12528.15) **[M1]**

12528
12528 **[A1]**

(b) $24000 \times 0.85^t = 9000$ or $0.85^t = 0.375$ **[M1]**

$t = \frac{\log 0.375}{\log 0.85}$ **[M1]**

6.04 **[A1]**

(c) $1.068^n = 2$ (or $1.068^n > 2$) **[M1]**

$n = \frac{\log 2}{\log 1.068} = 10.536\ldots$ **[M1]**

11 **[A1]**

(d) $2^y = \frac{x-3}{5}$ **[M1]**

$y = \log_2\left(\frac{x-3}{5}\right)$ or $y \log 2 = \log\left(\frac{x-3}{5}\right)$ **[M1]**

$log_2(\frac{x-3}{5}$ or
log
(
3
5
)

log
2
log2
log(
5
x3
)

[A1]

[Back to Question] *[Total Marks: 11]*

39 **(a)** Substitutes $(0, 7)$ to find A: $7 = A \times b^0 \implies A = 7$ **[B1]**

Substitutes $(3, 56)$ with $A = 7$: $7 \times b^3 = 56 \implies b^3 = 8$ **[M1]**

$b = 2$ **[A1]**

(b) Sets $y = 7 \times 2^x$ **[M1]**

Rearranges to make 2^x the subject: $2^x = \dfrac{y}{7}$ **[M1]**

Converts to logarithmic form: $\log_2\left(\dfrac{x}{7}\right)$ (or equivalent correct form) **[A1]**

(c) **(i)** $\dfrac{600}{7}$ **[B1]**

(ii) $x = \dfrac{\log(600/7)}{\log 2}$ **[M1]**

6.421 (from 6.4214...) **[A1]**

(d) **(i)** Sets $4 = \log_c 81 \implies c^4 = 81$ **[M1]**

$c = 3$ **[A1]**

(ii) $\dfrac{1}{9}$ (or 3^{-2}) **[B1]**

[Back to Question] *[Total Marks: 12]*